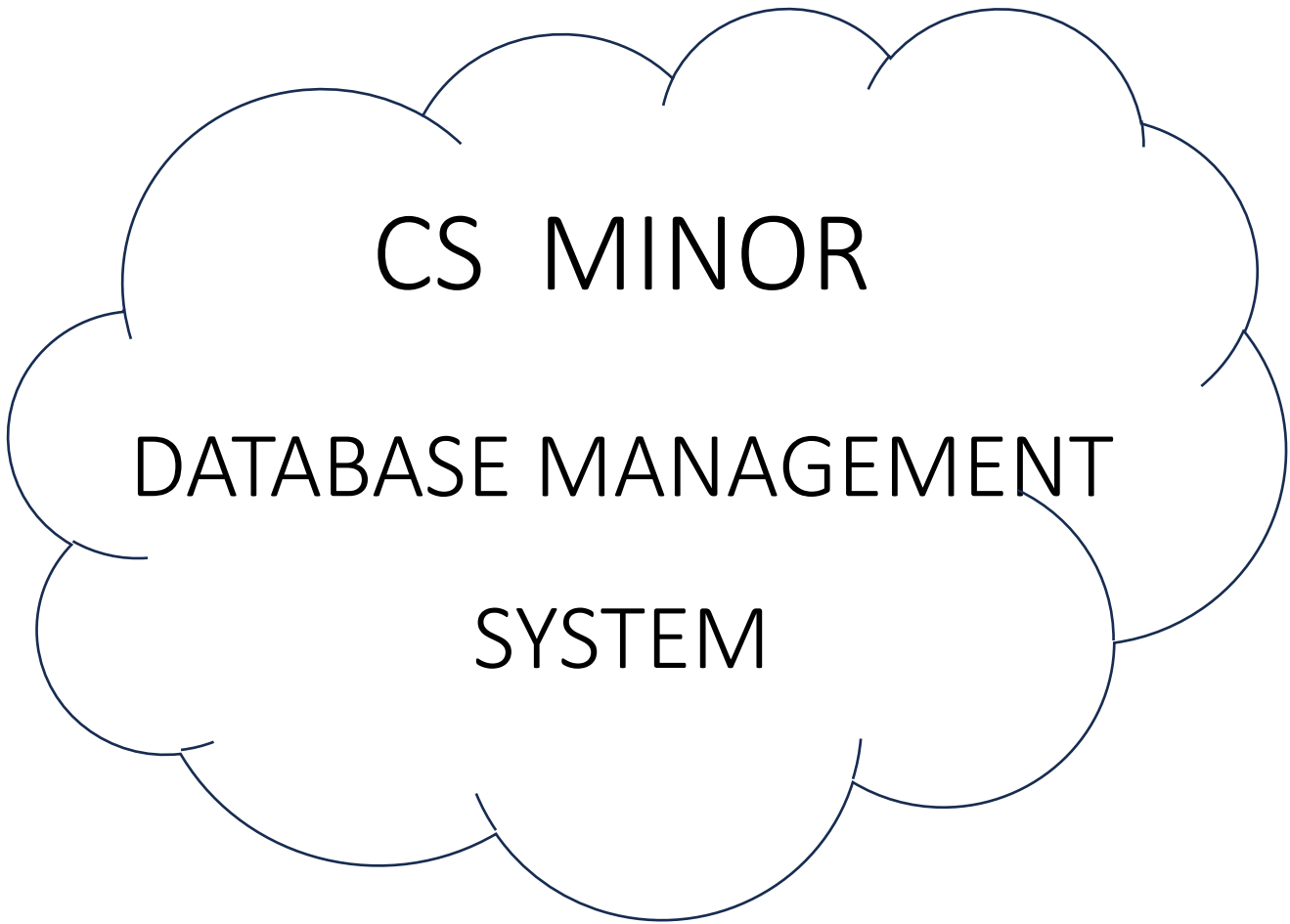




CS MINOR DATABASE MANAGEMENT SYSTEM

LIBRARY MANAGEMENT SYSTEM

SUBMITTED BY:

SETHURAMAN M (2201108047)

SANTTOSH A (2201115060)

SHRINITHI S (2201110054)

SUBMITTED TO:

DR. K. SARULADHA

TYPES OF CARDINALITY USED

S.No.	ENTITIES	TYPE OF CARDINALITY
1.	LIBRARY - BOOK	ONE TO MANY
2.	LIBRARY TO ADMIN	ONE TO ONE
3.	LIBRARY TO BOOK RECORD	ONE TO MANY
4.	LIBRARY TO AUTHENTICATION SYSTEM	ONE TO MANY
5.	BOOK TO AUTHOR	ONE TO MANY
6.	BOOK TO USER	MANY TO MANY
7.	USER TO TRANSACTION	ONE TO MANY
8.	BOOK TO PUBLISHER	ONE TO ONE

TYPES OF ATTRIBUTES USED

S.No.	TYPES	ATTRIBUTES
1.	SIMPLE	YEAR OF PUBLISHING, AUTHOR_QUALIFICATION, AUTHOR_CATEGORY, PUBLISHER_NAME, NO_OF_COPIES, PASSWORD, DOB, NO_OF_BOOKS
2.	COMPOSITE	AUTHOR_NAME, PUBLISHER_ADDRESS, ADMIN_NAME, USER_NAME
3.	SINGLE VALUED	BOOK_NAME, PRICE, EDITION, CATEGORY, CONTACT_NUMBER(PUBLISHER & USER), STATUS, ISSUE_DATE, RETURN_DATE
4.	MULTI VALUED	AUTHOR_ID, BOOK_ID, RETURN_STATUS, FINE_AMT, DUE_FEES
5.	DERIVED	RETURN_STATUS, FINE_AMT, DUE_FEES, TRANSACTION_ID
6.	KEY	BOOK_ID, AUTHOR_ID, PUBLISHER_ID, ADMIN_ID, LOGIN_ID, RECORD_ID, TRANSACTION_ID, USER_ID
7.	NULL	RETURN_DATE
8.	FOREIGN	AUTHOR_ID_1, AUTHOR_ID_2, BOOK_ID, USER_ID(BOOK RECORD),

BOOK:

Book_id	Book_name	Author_id_1	Author_id_2	Price	Edition	Category	Year_of_publish	Return_status
B001	Mystery of India	A001	A002	250	1st	Fiction	2010	Returned
B002	The Indian Saga	A003	A004	300	2nd	History	1990	Not returned
B003	Science Simplified	A005	A006	350	1st	Science	2015	Returned
B004	Art of Living	A007	A008	400	3rd	Self-Help	2011	Not returned
B005	Digital Revolution	A009	A010	500	1st	Technology	1989	Not returned
B006	Indian Cuisine	A011	A012	200	1st	Cooking	1992	Returned
B007	Bollywood Beats	A013	A014	150	2nd	Entertainment	2013	Returned
B008	Yoga for Life	A015	A016	220	1st	Health	2016	Not returned
B009	Economic Growth	A017	A018	330	2nd	Economics	1985	Returned
B010	Spiritual Journey	A019	A020	280	1st	Religion	2011	Returned
B011	Ancient India	A021	A022	260	1st	History	2008	Available
B012	Modern India	A023	A024	310	2nd	History	2017	Available
B013	Tech Innovations	A025	A026	480	1st	Technology	2019	Available
B014	Health and Well...	A027	A028	290	1st	Health	2012	Available
B015	Indian Myths	A029	A030	260	1st	Fiction	2007	Available

AUTHOR:

Author_id	Book_id	Author_name	Qualification	Author_category
A001	B001	Rahul Sharma	PhD	Fiction
A002	B001	Priya Singh	M.A.	Fiction
A003	B002	Amit Verma	M.Sc	History
A004	B002	Sunita Rao	PhD	History
A005	B003	Vikram Joshi	B.Tech	Science
A006	B003	Neha Gupta	PhD	Science
A007	B004	Sanjay Mehta	MBA	Self-Help
A008	B004	Kavita Desai	PhD	Self-Help
A009	B005	Manish Kapoor	M.A.	Technology
A010	B005	Ritu Malhotra	B.Tech	Technology
A011	B006	Arun Patel	Diploma	Cooking
A012	B006	Sneha Shah	M.Sc	Cooking
A013	B007	Deepak Kumar	MBA	Entertainment
A014	B007	Anjali Menon	B.Com	Entertainment
A015	B008	Rohit Verma	B.A.	Health
A016	B008	Meera Nair	M.Sc	Health
A017	B009	Pankaj Singh	MBA	Economics
A018	B009	Divya Patel	PhD	Economics
A019	B010	Rajesh Kumar	B.Tech	Religion
A020	B010	Snehal Desai	M.A.	Religion
A021	B011	Suresh Reddy	PhD	History
A022	B011	Lakshmi Rao	M.A.	History
A023	B012	Vijay Singh	M.Sc	History
A024	B012	Anita Iyer	PhD	History
A025	B013	Sahil Gupta	B.Tech	Technology
A026	B013	Pooja Kapoor	MBA	Technology
A027	B014	Karan Mehta	B.Com	Health
A028	B014	Divya Sharma	M.Sc	Health
A029	B015	Anand Patel	M.A.	Fiction
A030	B015	Rekha Singh	PhD	Fiction

BOOK_RECORD:

Record_id	User_id	Book_id	Issue_date	Return_date	Return_status
R001	U001	B001	2021-01-01	2021-01-15	Returned
R002	U002	B001	2021-02-20	2021-03-05	Returned
R003	U003	B002	2021-01-10	2021-01-27	Returned
R004	U004	B003	2021-02-12	2021-02-28	Returned
R005	U005	B004	2021-03-05	NULL	Not Returned
R006	U006	B005	2021-03-15	NULL	Not Returned
R007	U007	B006	2021-01-18	2021-02-02	Returned
R008	U008	B007	2021-02-22	2021-03-10	Returned
R009	U009	B008	2021-03-01	NULL	Not Returned
R010	U010	B009	2021-03-03	2021-03-18	Returned
R011	U001	B010	2021-03-07	2021-03-22	Returned
R017	U007	B002	2021-04-05	NULL	Not Returned
R018	U008	B004	2021-04-10	NULL	Not Returned
R019	U009	B004	2021-04-15	NULL	Not Returned
R020	U010	B005	2021-04-20	NULL	Not Returned

TRANSACTION:

Transaction_id	User_id	Book_id_1	Return_status_1	Fine_amt_1	Due_fees_1	Book_id_2	Return_status_2	Fine_amt_2	Due_fees_2
T001	U001	B001	Returned	0	0	B010	Returned	10	0
T002	U006	B005	Not Returned	0	10	NULL	NULL	0	0
T003	U003	B002	Returned	30	0	NULL	NULL	0	0
T004	U002	B001	Returned	0	0	NULL	NULL	5	0
T005	U005	B004	Not Returned	5	20	NULL	NULL	0	0
T006	U004	B003	Returned	0	0	NULL	NULL	0	0
T007	U007	B006	Returned	0	0	B002	Not Returned	0	0
T008	U008	B007	Returned	30	0	B004	Not Returned	0	5
T009	U009	B008	Not Returned	0	15	B004	Not Returned	5	0
T010	U010	B009	Returned	0	0	B005	Not Returned	0	10

ADMIN:

Admin_id	Admin_name
ADM001	Ramesh Singh
ADM002	Suresh Patel
ADM003	Vijay Kumar
ADM004	Anita Desai
ADM005	Sunil Gupta
ADM006	Priya Sharma
ADM007	Rahul Verma
ADM008	Kavita Rao

AUTHENTICATION_SYSTEM:

Admin_id	Login_id	Password
ADM001	ramesh.s	pass123
ADM002	suresh.p	pass234
ADM003	vijay.k	pass345
ADM004	anita.d	pass456
ADM005	sunil.g	pass567
ADM006	priya.s	pass678
ADM007	rahul.v	pass789
ADM008	kavita.r	pass890

USER:

User_id	User_name	DOB	Contact_number	Book_id_1	Return_status_1	Book_id_2	Return_status_2	Number_of_books	Transaction_id
U001	Arjun Kumar	1990-05-12	9811122233	B001	Returned	B010	Returned	2	T001
U002	Bhavna Reddy	1988-08-23	9911122233	B001	Returned	NULL	NULL	1	T004
U003	Chandra Singh	1992-01-01	9819988776	B002	Returned	NULL	NULL	1	T003
U004	Deepa Iyer	1991-02-14	9919988776	B003	Returned	NULL	NULL	1	T006
U005	Esha Menon	1993-03-30	9812341234	B004	Not Returned	NULL	NULL	1	T005
U006	Farhan Khan	1987-04-21	9912341234	B005	Not Returned	NULL	NULL	1	T002
U007	Gauri Desai	1990-10-10	9811123344	B006	Returned	B002	Not Returned	2	T007
U008	Harish Patel	1989-05-05	9911123344	B007	Returned	B004	Not Returned	2	T008
U009	Isha Gupta	1995-07-17	9819988866	B008	Not Returned	B004	Not Returned	2	T009
U010	Jatin Kumar	1994-09-29	9919988866	B009	Returned	B005	Not Returned	2	T010

PUBLISHER:

Publisher_id	Publisher_name	Book_id	Book_name	Edition	Year_of_publish	Publisher_address	Contact_number	No_of_copies	Status
P001	PearsonIndia	B001	Mystery of India	1st	2010	Mumbai, India	9812345678	50	Active
P002	McGraw Hill India	B002	The Indian Saga	2nd	1990	Delhi, India	9912345678	40	Active
P003	PearsonIndia	B003	Science Simplified	1st	2015	Bangalore, India	9811122233	30	Active
P004	PearsonIndia	B004	Art of Living	3rd	2011	Chennai, India	9911122233	25	Active
P005	Oxford India	B005	Digital Revolution	1st	1989	Kolkata, India	9819988776	35	Active
P006	Cambridge India	B006	Indian Cuisine	1st	1992	Hyderabad, India	9919988776	20	Active
P007	Rupa Publications	B007	Bollywood Beats	2nd	2013	Mumbai, India	9812341234	45	Active
P008	Penguin India	B008	Yoga for Life	1st	2016	Delhi, India	9912341234	50	Active
P009	HarperCollins India	B009	Economic Growth	2nd	1985	Bangalore, India	9811123344	40	Active
P010	Rupa Publications	B010	Spiritual Journey	1st	2011	Chennai, India	9911123344	30	Active
P011	PearsonIndia	B011	Ancient India	1st	2008	Kolkata, India	9819988866	20	Active
P012	Cambridge India	B012	Modern India	2nd	2017	Hyderabad, India	9919988866	25	Active
P013	PearsonIndia	B013	Tech Innovations	1st	2019	Mumbai, India	9812345566	30	Active
P014	McGraw Hill India	B014	Health and Well...	1st	2012	Delhi, India	9912345566	35	Active
P015	Scholastic India	B015	Indian Myths	1st	2007	Bangalore, India	9811126677	40	Active

NORMALIZATION

1NF:

ALL THE TABLES GIVEN HERE ARE IN 1NF FORM. THE TABLES BOOK, AUTHOR, BOOK RECORD, TRANSACTIONS, USER, PUBLISHER ARE DECOMPOSED TO 1NF. FOR EXAMPLE, BOOK TABLE:

BEFORE DECOMPOSITION:

Book_id	Book_name	Authors	Price	Edition	Category	Year_of_publish	Return_status
B001	Mystery of India	A001, A002	250	1st	Fiction	2010	Returned
B002	The Indian Saga	A003, A004	300	2nd	History	1990	Not returned
B003	Science Simplified	A005, A006	350	1st	Science	2015	Returned
B004	Art of Living	A007, A008	400	3rd	Self-Help	2011	Not returned
B005	Digital Revolution	A009, A010	500	1st	Technology	1989	Not returned
B006	Indian Cuisine	A011, A012	200	1st	Cooking	1992	Returned
B007	Bollywood Beats	A013, A014	150	2nd	Entertainment	2013	Returned
B008	Yoga for Life	A015, A016	220	1st	Health	2016	Not returned
B009	Economic Growth	A017, A018	330	2nd	Economics	1985	Returned
B010	Spiritual Journey	A019, A020	280	1st	Religion	2011	Returned
B011	Ancient India	A021, A022	260	1st	History	2008	Available
B012	Modern India	A023, A024	310	2nd	History	2017	Available
B013	Tech Innovations	A025, A026	480	1st	Technology	2019	Available
B014	Health and Wellness	A027, A028	290	1st	Health	2012	Available
B015	Indian Myths	A029, A030	260	1st	Fiction	2007	Available

AFTER DECOMPOSITION:

Book_id	Book_name	Author_id_1	Author_id_2	Price	Edition	Category	Year_of_publish	Return_status
B001	Mystery of India	A001	A002	250	1st	Fiction	2010	Returned
B002	The Indian Saga	A003	A004	300	2nd	History	1990	Not returned
B003	Science Simplified	A005	A006	350	1st	Science	2015	Returned
B004	Art of Living	A007	A008	400	3rd	Self-Help	2011	Not returned
B005	Digital Revolution	A009	A010	500	1st	Technology	1989	Not returned
B006	Indian Cuisine	A011	A012	200	1st	Cooking	1992	Returned
B007	Bollywood Beats	A013	A014	150	2nd	Entertainment	2013	Returned
B008	Yoga for Life	A015	A016	220	1st	Health	2016	Not returned
B009	Economic Growth	A017	A018	330	2nd	Economics	1985	Returned
B010	Spiritual Journey	A019	A020	280	1st	Religion	2011	Returned
B011	Ancient India	A021	A022	260	1st	History	2008	Available
B012	Modern India	A023	A024	310	2nd	History	2017	Available
B013	Tech Innovations	A025	A026	480	1st	Technology	2019	Available
B014	Health and Well...	A027	A028	290	1st	Health	2012	Available
B015	Indian Myths	A029	A030	260	1st	Fiction	2007	Available

2NF:

HERE THE TABLES ADMIN, AUTHENTICATION_SYSTEM, BOOK_RECORD SATISFY 2NF.

PUBLISHER TABLE DOESN'T SATISFY 2NF. HENCE IT IS DECOMPOSED.

BEFORE DECOMPOSITION:

Publisher_id	Publisher_name	Book_id	Book_name	Edition	Year_of_publish	Publisher_address	Contact_number	No_of_copies	Status
P001	PearsonIndia	B001	Mystery of India	1st	2010	Mumbai, India	9812345678	50	Active
P002	McGraw Hill India	B002	The Indian Saga	2nd	1990	Delhi, India	9912345678	40	Active
P003	PearsonIndia	B003	Science Simplified	1st	2015	Bangalore, India	9811122233	30	Active
P004	PearsonIndia	B004	Art of Living	3rd	2011	Chennai, India	9911122233	25	Active
P005	Oxford India	B005	Digital Revolution	1st	1989	Kolkata, India	9819988776	35	Active
P006	Cambridge India	B006	Indian Cuisine	1st	1992	Hyderabad, India	9919988776	20	Active
P007	Rupa Publications	B007	Bollywood Beats	2nd	2013	Mumbai, India	9812341234	45	Active
P008	Penguin India	B008	Yoga for Life	1st	2016	Delhi, India	9912341234	50	Active
P009	HarperCollins India	B009	Economic Growth	2nd	1985	Bangalore, India	9811123344	40	Active
P010	Rupa Publications	B010	Spiritual Journey	1st	2011	Chennai, India	9911123344	30	Active
P011	PearsonIndia	B011	Ancient India	1st	2008	Kolkata, India	9819988866	20	Active
P012	Cambridge India	B012	Modern India	2nd	2017	Hyderabad, India	9919988866	25	Active
P013	PearsonIndia	B013	Tech Innovations	1st	2019	Mumbai, India	9812345566	30	Active
P014	McGraw Hill India	B014	Health and Well...	1st	2012	Delhi, India	9912345566	35	Active
P015	Scholastic India	B015	Indian Myths	1st	2007	Bangalore, India	9811126677	40	Active

AFTER DECOMPOSITION:

PUBLISHER_INFO:

Publisher_id	Publisher_name	Publisher_address	Contact_number
P001	PearsonIndia	Mumbai, India	9812345678
P002	McGraw Hill India	Delhi, India	9912345678
P003	PearsonIndia	Bangalore, India	9811122233
P004	PearsonIndia	Chennai, India	9811122233
P005	Oxford India	Kolkata, India	9819988776
P006	Cambridge India	Hyderabad, India	9919988776
P007	Rupa Publications	Mumbai, India	9912341234
P008	Penguin India	Delhi, India	9912341234
P009	HarperCollins India	Bangalore, India	9811123344
P010	Rupa Publications	Chennai, India	9911123344
P011	PearsonIndia	Kolkata, India	9819988866
P012	Cambridge India	Hyderabad, India	9819988866
P013	PearsonIndia	Mumbai, India	9812345566
P014	McGraw Hill India	Delhi, India	9912345566
P015	Scholastic India	Bangalore, India	9811126677

PUBLISHER_BOOK:

Book_id	Book_name	Edition	Year_of_publish	No_of_copies	Status	Publisher_id
B001	Mystery of India	1st	2010	50	Active	P001
B002	The Indian Saga	2nd	1999	40	Active	P002
B003	Science Simplified	1st	2015	30	Active	P003
B004	Art of Living	3rd	2011	25	Active	P004
B005	Digital Revolution	1st	2010	35	Active	P005
B006	Indian Cuisine	1st	1992	20	Active	P006
B007	Bollywood Beats	1st	2013	45	Active	P007
B008	Yoga for Life	1st	2016	30	Active	P008
B009	Economic Growth	2nd	1985	40	Active	P009
B010	Spiritual Journey	1st	2011	30	Active	P010
B011	Ancient India	1st	2008	20	Active	P011
B012	Modern India	2nd	2017	25	Active	P012
B013	Tech Innovations	1st	2019	30	Active	P013
B014	Health and Wellness	1st	2012	35	Active	P014
B015	Indian Myths	1st	2007	40	Active	P015

3NF:

THE DECOMPOSED PUBLISHER TABLE(PUBLISHER_INFO, PUBLISHER_BOOK) CAN BE FURTHER DIVIDED FOR 3NF.

PUBLISHER_INFO CAN BE DIVIDED INTO PUBLISHER_INFORMATION, PUBLISHER_CONTACT.

PUBLISHER_INFORMATION:

Publisher_id	Publisher_name	Publisher_address
P001	PearsonIndia	Mumbai, India
P002	McGraw Hill India	Delhi, India
P003	PearsonIndia	Bangalore, India
P004	PearsonIndia	Chennai, India
P005	Oxford India	Kolkata, India
P006	Cambridge India	Hyderabad, India
P007	Rupa Publications	Mumbai, India
P008	Penguin India	Delhi, India
P009	HarperCollins India	Bangalore, India
P010	Rupa Publications	Chennai, India
P011	PearsonIndia	Kolkata, India
P012	Cambridge India	Hyderabad, India
P013	PearsonIndia	Mumbai, India
P014	McGraw Hill India	Delhi, India
P015	Scholastic India	Bangalore, India

PUBLISHER_CONTACT:

Publisher_id	Contact_number
P001	9812345678
P002	9912345678
P003	9811122233
P004	9811122233
P005	9819988776
P006	9919988776
P007	9912341234
P008	9912341234
P009	9811123344
P010	9911123344
P011	9819988866
P012	9819988866
P013	9812345566
P014	9912345566
P015	9811126677

PUBLISHER_BOOK CAN BE DIVIDED INTO PUBLISHER_BOOK_INFO, PUBLISHER_STATUS

PUBLISHER_BOOK_INFO:

Book_id	Book_name	Edition	Year_of_publish	No_of_copies	Publisher_id
B001	Mystery of India	1st	2010	50	P001
B002	The Indian Saga	2nd	1999	40	P002
B003	Science Simplified	1st	2015	30	P003
B004	Art of Living	3rd	2011	25	P004
B005	Digital Revolution	1st	2010	35	P005
B006	Indian Cuisine	1st	1992	20	P006
B007	Bollywood Beats	1st	2013	45	P007
B008	Yoga for Life	1st	2016	30	P008
B009	Economic Growth	2nd	1985	40	P009
B010	Spiritual Journey	1st	2011	30	P010
B011	Ancient India	1st	2008	20	P011
B012	Modern India	2nd	2017	25	P012
B013	Tech Innovations	1st	2019	30	P013
B014	Health and Wellness	1st	2012	35	P014
B015	Indian Myths	1st	2007	40	P015

BOOK_STATUS:

Book_id	No_of_copies	Status
B001	50	Active
B002	40	Active
B003	30	Active
B004	25	Active
B005	35	Active
B006	20	Active
B007	45	Active
B008	30	Active
B009	40	Active
B010	30	Active
B011	20	Active
B012	25	Active
B013	30	Active
B014	35	Active
B015	40	Active

SQL QUERIES:

1. Find books that were borrowed but never returned, along with user details.

```
SELECT U.User_id, U.User_name, B.Book_id, B.Book_name, BR.Issue_date  
FROM Book_record BR  
JOIN User U ON BR.User_id = U.User_id  
JOIN Book B ON BR.Book_id = B.Book_id  
WHERE BR.Issue_date IS NOT NULL  
AND BR.Return_date IS NULL;
```

User_id	User_name	Book_id	Book_name	Issue_date
U005	Esha Menon	B004	Art of Living	2021-03-05
U006	Farhan Khan	B005	Digital Revolution	2021-03-15
U007	Gauri Desai	B002	The Indian Saga	2021-04-05
U008	Harish Patel	B004	Art of Living	2021-04-10
U009	Isha Gupta	B004	Art of Living	2021-04-15
U009	Isha Gupta	B008	Yoga for Life	2021-03-01
U010	Jatin Kumar	B005	Digital Revolution	2021-04-20

2. Generate a report of all transactions with user, book, author, and publisher details.

```
SELECT T.Transaction_id, U.User_id, U.User_name AS name, B.Book_id,  
B.Book_name, P.Publisher_name
```

```
FROM transactions T
```

```
JOIN User U ON T.User_id = U.User_id
```

```
JOIN Book B ON T.Book_id_1 = B.Book_id
```

```
JOIN Author A ON B.Book_id = A.Book_id
```

```
JOIN Publisher P ON B.Book_id = P.Book_id
```

```
UNION
```

```
SELECT T.Transaction_id, U.User_id, U.User_name AS name, B.Book_id,  
B.Book_name, P.Publisher_name
```

```
FROM transactions T
```

```
JOIN User U ON T.User_id = U.User_id
```

```
JOIN Book B ON T.Book_id_2 = B.Book_id
```

```
JOIN Author A ON B.Book_id = A.Book_id
```

```
JOIN Publisher P ON B.Book_id = P.Book_id
```

```
ORDER BY name;
```

Transaction_id	User_id	name	Book_id	Book_name	Publisher_name
T001	U001	Arjun Kumar	B001	Mystery of India	PearsonIndia
T001	U001	Arjun Kumar	B010	Spiritual Journey	Rupa Publications
T004	U002	Bhavna Reddy	B001	Mystery of India	PearsonIndia
T003	U003	Chandra Singh	B002	The Indian Saga	McGraw Hill India
T006	U004	Deepa Iyer	B003	Science Simplified	PearsonIndia
T005	U005	Esha Menon	B004	Art of Living	PearsonIndia
T002	U006	Farhan Khan	B005	Digital Revolution	Oxford India
T007	U007	Gauri Desai	B006	Indian Cuisine	Cambridge India
T007	U007	Gauri Desai	B002	The Indian Saga	McGraw Hill India
T008	U008	Harish Patel	B007	Bollywood Beats	Rupa Publications
T008	U008	Harish Patel	B004	Art of Living	PearsonIndia
T009	U009	Isha Gupta	B008	Yoga for Life	Penguin India
T009	U009	Isha Gupta	B004	Art of Living	PearsonIndia
T010	U010	Jatin Kumar	B009	Economic Growth	HarperCollins India
T010	U010	Jatin Kumar	B005	Digital Revolution	Oxford India

3. Find books that were published before the year 2000 but are still being borrowed.

```
SELECT DISTINCT B.Book_id, B.Book_name, B.Year_of_publish
FROM Book B
JOIN Book_record BR ON B.Book_id = BR.Book_id
WHERE B.Year_of_publish < 2000
AND BR.Issue_date IS NOT NULL
AND BR.Return_date IS NULL;
```

Book_id	Book_name	Year_of_publish
B005	Digital Revolution	1989
B002	The Indian Saga	1990

4. Find the publisher with the highest number of published books.

```
SELECT Publisher_name, COUNT(Book_id) AS published_count
FROM Publisher
GROUP BY Publisher_name
ORDER BY published_count DESC
LIMIT 1
```

Publisher_name	published_count
PearsonIndia	5

5. Find the most borrowed book in the library

```
SELECT B.Book_id, B.Book_name, COUNT(*) AS borrow_count
FROM Book_record BR
JOIN book B ON B.Book_id=BR.Book_id
GROUP BY Book_id
ORDER BY borrow_count DESC
LIMIT 1
```

Book_id	Book_name	borrow_count
B004	Art of Living	3

6. Find the top 3 users who have paid the highest fines.

```
SELECT User_id, SUM(Fine_amt_1 + Fine_amt_2 + Fine_amt_3) AS total_fine  
FROM transactions  
GROUP BY User_id  
ORDER BY total_fine DESC  
LIMIT 3
```

User_id	total_fine
U008	30
U003	30
U001	10

7. Find the earliest and latest book published in the system

```
SELECT Book_Name, Year_of_publish  
FROM Book  
WHERE Year_of_publish = (SELECT MIN(Year_of_publish) FROM Book)  
OR Year_of_publish = (SELECT MAX(Year_of_publish) FROM Book);
```

Book_Name	Year_of_publish
Economic Growth	1985
Tech Innovations	2019

8. Find the most recently borrowed book

```
SELECT Br.Book_id, Book_name, Issue_date  
FROM Book_record Br JOIN book B on Br.Book_id=B.Book_id  
WHERE Issue_date = (SELECT MAX(Issue_date) FROM Book_record);
```

Book_id	Book_name	Issue_date
B005	Digital Revolution	2021-04-20

9.TRIGGER: AUTOMATICALLY UPDATE FINE WHEN A BOOK IS RETURNED LATE

```
CREATE TRIGGER trg_update_fine
BEFORE UPDATE ON book_record
FOR EACH ROW
BEGIN
    DECLARE L_DUE_DATE DATE;
    DECLARE L_DAYS_LATE INT;
    IF NEW.return_date IS NOT NULL THEN
        SET L_DUE_DATE = DATE_ADD(NEW.issue_date, INTERVAL 14 DAY);
    IF NEW.return_date > L_DUE_DATE THEN
        SET L_DAYS_LATE = DATEDIFF(NEW.return_date, L_DUE_DATE);
        UPDATE transactions
        SET FINE_AMT_1 = COALESCE(FINE_AMT_1, 0) + (L_DAYS_LATE * 10)
        WHERE BOOK_ID_1 = NEW.BOOK_ID;
    END IF;
    SET NEW.RETURN_STATUS = 'RETURNED';
END IF;
END
BEFORE EXECUTION:
```

Transaction_id	User_id	Book_id_1	Return_status_1	Fine_amt_1	Due_fees_1	Book_id_2	Return_status_2	Fine_amt_2	Due_fees_2
T001	U001	B001	Returned	0	0	B010	Returned	10	0
T002	U006	B005	Not Returned	0	10	NULL	NULL	0	0
T003	U003	B002	Not Returned	0	0	NULL	NULL	0	0
T004	U002	B001	Returned	0	0	NULL	NULL	5	0
T005	U005	B004	Not Returned	5	20	NULL	NULL	0	0
T006	U004	B003	Returned	0	0	NULL	NULL	0	0
T007	U007	B006	Returned	0	0	B002	Not Returned	0	0
T008	U008	B007	Returned	30	0	B004	Not Returned	0	5
T009	U009	B008	Not Returned	0	15	B004	Not Returned	5	0
T010	U010	B009	Returned	0	0	B005	Not Returned	0	10

Record_id	User_id	Book_id	Issue_date	Return_date	Return_status
R001	U001	B001	2021-01-01	2021-01-15	Returned
R002	U002	B001	2021-02-20	2021-03-05	Returned
R003	U003	B002	2021-01-10	NULL	Not Returned
R004	U004	B003	2021-02-12	2021-02-28	Returned
R005	U005	B004	2021-03-05	NULL	Not Returned
R006	U006	B005	2021-03-15	NULL	Not Returned
R007	U007	B006	2021-01-18	2021-02-02	Returned
R008	U008	B007	2021-02-22	2021-03-10	Returned
R009	U009	B008	2021-03-01	NULL	Not Returned
R010	U010	B009	2021-03-03	2021-03-18	Returned
R011	U001	B010	2021-03-07	2021-03-22	Returned
R017	U007	B002	2021-04-05	NULL	Not Returned
R018	U008	B004	2021-04-10	NULL	Not Returned
R019	U009	B004	2021-04-15	NULL	Not Returned
R020	U010	B005	2021-04-20	NULL	Not Returned

User_id	User_name	DOB	Contact_number	Book_id_1	Return_status_1	Book_id_2	Return_status_2	Number_of_books	Transaction_id
U001	Arjun Kumar	1990-05-12	9811122233	B001	Returned	B010	Returned	2	T001
U002	Bhavna Reddy	1988-08-23	9911122233	B001	Returned	NULL	NULL	1	T004
U003	Chandra Singh	1992-01-01	9819988776	B002	Not Returned	NULL	NULL	1	T003
U004	Deepa Iyer	1991-02-14	9919988776	B003	Returned	NULL	NULL	1	T006
U005	Esha Menon	1993-03-30	9812341234	B004	Not Returned	NULL	NULL	1	T005
U006	Farhan Khan	1987-04-21	9912341234	B005	Not Returned	NULL	NULL	1	T002
U007	Gauri Desai	1990-10-10	9811123344	B006	Returned	B002	Not Returned	2	T007
U008	Harish Patel	1989-05-05	9911123344	B007	Returned	B004	Not Returned	2	T008
U009	Isha Gupta	1995-07-17	9819988866	B008	Not Returned	B004	Not Returned	2	T009
U010	Jatin Kumar	1994-09-29	9919988866	B009	Returned	B005	Not Returned	2	T010

AFTER EXECUTION:

Record_id	User_id	Book_id	Issue_date	Return_date	Return_status
R001	U001	B001	2021-01-01	2021-01-15	Returned
R002	U002	B001	2021-02-20	2021-03-05	Returned
R003	U003	B002	2021-01-10	2021-01-28	Returned
R004	U004	B003	2021-02-12	2021-02-28	Returned
R005	U005	B004	2021-03-05	NULL	Not Returned
R006	U006	B005	2021-03-15	NULL	Not Returned
R007	U007	B006	2021-01-18	2021-02-02	Returned
R008	U008	B007	2021-02-22	2021-03-10	Returned
R009	U009	B008	2021-03-01	NULL	Not Returned
R010	U010	B009	2021-03-03	2021-03-18	Returned
R011	U001	B010	2021-03-07	2021-03-22	Returned
R017	U007	B002	2021-04-05	NULL	Not Returned
R018	U008	B004	2021-04-10	NULL	Not Returned
R019	U009	B004	2021-04-15	NULL	Not Returned
R020	U010	B005	2021-04-20	NULL	Not Returned

Transaction_id	User_id	Book_id_1	Return_status_1	Fine_amt_1	Due_fees_1	Book_id_2	Return_status_2	Fine_amt_2	Due_fees_2
T001	U001	B001	Returned	0	0	B010	Returned	10	0
T002	U006	B005	Not Returned	0	10	NULL	NULL	0	0
T003	U003	B002	Returned	30	0	NULL	NULL	0	0
T004	U002	B001	Returned	0	0	NULL	NULL	5	0
T005	U005	B004	Not Returned	5	20	NULL	NULL	0	0
T006	U004	B003	Returned	0	0	NULL	NULL	0	0
T007	U007	B006	Returned	0	0	B002	Not Returned	0	0
T008	U008	B007	Returned	30	0	B004	Not Returned	0	5
T009	U009	B008	Not Returned	0	15	B004	Not Returned	5	0
T010	U010	B009	Returned	0	0	B005	Not Returned	0	10

User_id	User_name	DOB	Contact_number	Book_id_1	Return_status_1	Book_id_2	Return_status_2	Number_of_books	Transaction_id
U001	Arjun Kumar	1990-05-12	9811122233	B001	Returned	B010	Returned	2	T001
U002	Bhavna Reddy	1988-08-23	9911122233	B001	Returned	NULL	NULL	1	T004
U003	Chandra Singh	1992-01-01	9819988776	B002	Returned	NULL	NULL	1	T003
U004	Deepa Iyer	1991-02-14	9919988776	B003	Returned	NULL	NULL	1	T006
U005	Esha Menon	1993-03-30	9812341234	B004	Not Returned	NULL	NULL	1	T005
U006	Farhan Khan	1987-04-21	9912341234	B005	Not Returned	NULL	NULL	1	T002
U007	Gauri Desai	1990-10-10	9811123344	B006	Returned	B002	Not Returned	2	T007
U008	Harish Patel	1989-05-05	9911123344	B007	Returned	B004	Not Returned	2	T008
U009	Isha Gupta	1995-07-17	9819988866	B008	Not Returned	B004	Not Returned	2	T009
U010	Jatin Kumar	1994-09-29	9919988866	B009	Returned	B005	Not Returned	2	T010

10. FUNCTION: GET TOTAL BOOKS BORROWED BY A USER

DELIMITER \$\$

CREATE FUNCTION GET_TOTAL_BOOKS_BORROWED(P_USER_ID VARCHAR(10))

RETURNS INT

DETERMINISTIC

BEGIN

DECLARE TOTAL_BOOKS INT;

SELECT COUNT(*)

INTO TOTAL_BOOKS

FROM BOOK_RECORD

WHERE USER_ID = P_USER_ID;

RETURN TOTAL_BOOKS;

END\$\$

DELIMITER ;

```
1 SELECT get_total_books_borrowed('U001') AS TotalBooksBorrowed;
2
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
TotalBooksBorrowed			
2			

11. PROCEDURE: GENERATE A USER REPORT (BORROWED BOOKS & TOTAL FINE)

DELIMITER \$\$

```
CREATE PROCEDURE GENERATE_USER_REPORT (IN P_USER_ID VARCHAR(10))
```

```
BEGIN
```

```
    DECLARE TOTAL_BOOKS INT DEFAULT 0;
```

```
    DECLARE TOTAL_FINE DECIMAL(10,2) DEFAULT 0;
```

```
    -- GET TOTAL BOOKS BORROWED
```

```
    SELECT COUNT(*)
```

```
        INTO TOTAL_BOOKS
```

```
        FROM BOOK_RECORD
```

```
        WHERE USER_ID = P_USER_ID;
```

```
    -- SUM THE FINES (ASSUMING FINES ARE STORED AS FINE_AMT_1,  
    FINE_AMT_2, FINE_AMT_3)
```

```
    SELECT IFNULL(SUM(FINE_AMT_1 + FINE_AMT_2), 0)
```

```
        INTO TOTAL_FINE
```

```
        FROM TRANSACTIONS
```

```
        WHERE USER_ID = P_USER_ID;
```

```
    -- REPORT OUTPUT
```

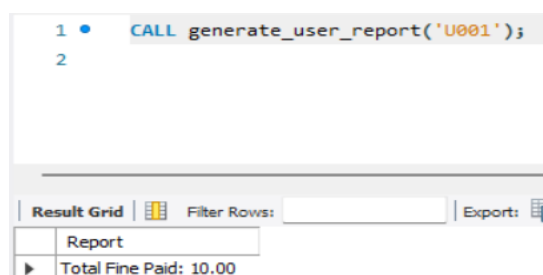
```
    SELECT CONCAT('USER ID: ', P_USER_ID) AS REPORT;
```

```
    SELECT CONCAT('TOTAL BOOKS BORROWED: ', TOTAL_BOOKS) AS REPORT;
```

```
    SELECT CONCAT('TOTAL FINE PAID: ', TOTAL_FINE) AS REPORT;
```

```
END$$
```

DELIMITER ;



12.SIMPLER CURSOR: LIST OVERDUE BOOKS THAT HAVE NOT BEEN RETURNED

DELIMITER \$\$

CREATE PROCEDURE LIST_OVERDUE_BOOKS()

BEGIN

DECLARE DONE INT DEFAULT 0;

DECLARE OVERDUE_BOOK_ID VARCHAR(10);

DECLARE OVERDUE_USER_ID VARCHAR(10);

DECLARE OVERDUE_ISSUE_DATE DATE;

DECLARE CUR CURSOR FOR

SELECT BOOK_ID, USER_ID, ISSUE_DATE

FROM BOOK_RECORD

WHERE RETURN_DATE IS NULL

AND RETURN_STATUS = 'NOT RETURNED'

AND DATE_ADD(ISSUE_DATE, INTERVAL 14 DAY) < CURDATE();

DECLARE CONTINUE HANDLER FOR NOT FOUND SET DONE = 1;

OPEN CUR;

READ_LOOP: LOOP

FETCH CUR INTO OVERDUE_BOOK_ID, OVERDUE_USER_ID,
OVERDUE_ISSUE_DATE;

IF DONE THEN

LEAVE READ_LOOP;

END IF;

SELECT CONCAT('BOOK ID: ', OVERDUE_BOOK_ID, ', USER ID: ',
OVERDUE_USER_ID, ', ISSUE DATE: ', DATE_FORMAT(OVERDUE_ISSUE_DATE,
'%Y-%M-%D')) AS OVERDUE_INFO;

END LOOP;

```
CLOSE CUR;  
END$$  
DELIMITER ;
```

```
1  CALL list_overdue_books();  
2  
3
```

Result Grid		Filter Rows:	Export
	overdue_info		
▶	Book ID: B004, User ID: U009, Issue Date: 2021-04-15		